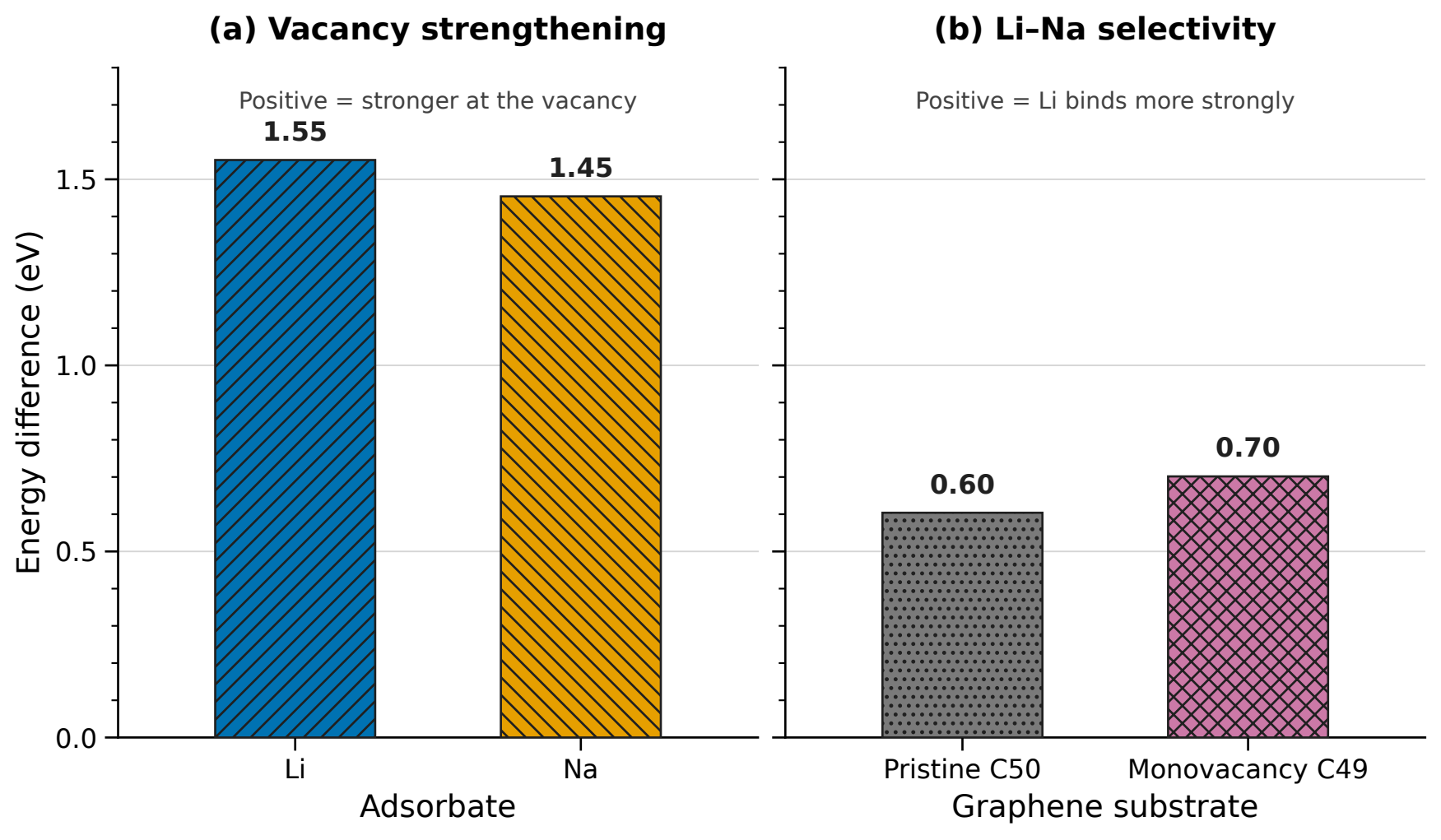

(a) Vacancy strengthening
Positive = stronger at the vacancy
1.55
1.45
Energy difference (eV)
1.5
1.0
0.5
0.0
Li
Na
Adsorbate
(b) Li–Na selectivity
Positive = Li binds more strongly
0.60
0.70
Pristine C50
Monovacancy C49
Graphene substrate